

2026.4.21. ODE III.

1. 特征方程方法.

对于常系数高阶线性 ODE.

如 $y'' - 3y' + 2y = 0$, $y^{(4)} - 3y''' + 2y'' - y' - y = 0$ 等.
使用特征方程求解.

eg. 以上 2 个方程分别对应特征方程

$$x^2 - 3x + 2 = 0.$$

$$x^4 - 3x^3 + 2x^2 - x - 1 = 0.$$

对于特征方程的一个 m 重特征根 α , 对应原方程
有形如 $\underbrace{p_m(x)}_{\text{虚数}} e^{\alpha x}$ 的基解. 其中 $p_m(x)$ 是任意 $\leq m$ 次多项式.
 α 导致三角函数. $e^{i\theta} = \cos\theta + i\sin\theta$.

变式 1. 非齐次. 如 $y'' - 3y' + 2y = f(x)$.

Step 1. 猜出一个特解. y^*

Step 2. 解对应齐次方程通解 $C_1 e^x + C_2 e^{2x}$

非齐次方程通解为 $f(x) + C_1 e^x + C_2 e^{2x}$.

① $f(x)$ 是多项式、指数、三角函数时, 直接猜

(i) $f(x) = A e^{\alpha x}$. 猜 $y = \underbrace{p_m(x) e^{\alpha x}}_{m=0 \text{ 即不是特征根}}$. (α 是 m 重特征根)

(ii) $f(x) = A \sin \beta x + B \cos \beta x$. 猜 $y = p_m(x) \sin \beta x + q_m(x) \cos \beta x$
($i\beta$ 是 m 重特征根).

(iii) $f(x) = p_m(x)$. m : 次多项式. 猜 $y = q_n(x)$. 取 n 使最高
次项对齐次数.

Ex. 对以下方程确定特解形式. (习题 9.5.3).

(1) $y'' - 5y' + 5y = 6$.

$$y = C \quad \leadsto \quad y = 6.$$

(2) $y'' + 3y' = 6e^{2x}$.

$$y = Ae^{2x} \quad \leadsto \quad y = \frac{3}{5}e^{2x}$$

(3) $y'' - 9y' + 20y = x + 1$

$$y = Ax + B \quad \leadsto \quad y = \frac{1}{20}x + \frac{29}{400}.$$

(4) $y'' + y = 4\sin x$. 注意 i 是 **一重特征根**.

$$y = (Ax+B)\sin x + (Cx+D)\cos x$$

(5) $y'' - 3y' + 2y = x\cos x$. i 不是特征根, 但右侧多了 x .

$$y = (Ax+B)\sin x + (Cx+D)\cos x.$$

(6) $y'' - 9y = \underline{e^{3x}\cos x}$. $\rightarrow$ 这对应 $e^{(3+i)x}, e^{(3-i)x}$.
 $3 \pm i$ 不是特征根.

$$y = Ae^{(3+i)x} + Be^{(3-i)x} \quad \text{或写成} \quad y = e^{3x}(A\sin x + B\cos x).$$

(7) $y'' - y = \underline{2e^x - x^2}$ 根据线性性质, 这可分成算 2 个特解

$$y'' - y = 2e^x. \quad \leadsto \quad y = (Ax+B)e^x.$$

$$y'' - y = -x^2 \quad \leadsto \quad y = Ax + Bx + C.$$

(8) $y'' + y' = \sin 4x - 2\sin 2x$

$$y'' + y' = \sin 4x \quad \leadsto \quad y = \cancel{Ax+B} A\sin 4x + B\cos 4x$$

$$y'' + y' = -2\sin 2x \quad \leadsto \quad y = A\sin 2x + B\cos 2x.$$

(习题 9.5.4)

$$(1) y'' + y = x^2 + x.$$

$$y = Ax^2 + Bx + C \leadsto y = x^2 + x - 2.$$

$$(2) y'' + y' = x - 2. \quad \text{注意左侧最高次是 } y'. \\ \text{因此仍待定二次函数!}$$

$$y = Ax^2 + Bx + C.$$

$$\text{或 } y' = Ax + B \leadsto y' = x - 3 \leadsto y = \frac{x^2}{2} - 3x + C.$$

$$(3) y'' + y = \underbrace{e^{3x}}_{\substack{\downarrow \\ 3 \text{ 不是特征根}}} \underbrace{(x-2)}_{\substack{\text{多一次} \\ > 0+1=1}} \quad y = (Ax+B)e^{3x}.$$

$$(4) y'' - y = \underbrace{e^x}_{\substack{\downarrow \\ 1 \text{ 是重根} \\ x e^x}} \underbrace{(x^2-1)}_{\substack{\rightarrow 2 \text{ 次} \\ 2+1=3}} \quad y = x e^x (Ax^2 + Bx + C)$$

$$(5) y''' + 3y'' + 3y' + y = \underbrace{e^{-x}}_{\substack{\downarrow \\ -1 \text{ 三重根}}} \underbrace{(x-5)}_{\substack{\rightarrow 1 \text{ 次}}} \quad y = x^3 e^{-x} (Ax+B)$$

$$(6) y'' - 2y' + 2y = e^x + x \cos x.$$

$$y'' - 2y' + 2y = e^x \leadsto y = A e^x \leadsto y = e^x.$$

$$y'' - 2y' + 2y = x \cos x \leadsto y = (Ax+B) \sin x + (Cx+D) \cos x.$$

$$\text{Eg. (2024.6). } y'' - 2y' + y = e^x.$$

Step 1. 猜 $y = A x^2 e^x$. (1 是二重根)

$$y' = A e^x (x^2 + 2x), \quad y'' = A e^x (x^2 + 4x + 2)$$

$$y'' - 2y' + y = 2A e^x \quad A = \frac{1}{2}.$$

Step 2. 解齐次方程通解. $y'' - 2y' + y = 0.$

$$y = C_1 e^x + C_2 x e^x.$$

$$\Rightarrow y = \left(\frac{1}{2}x^2 + Ax + B\right) e^x.$$

Ex. (2023.3) $y'' + y' - 2y = x + e^x + \sin x$. $y(0) = -\frac{7}{20}$, $y'(0) = \frac{38}{15}$.

$x^2 + x - 2 = (x+2)(x-1)$. 特征根 $-2, 1$

Step 1. 特解

$y'' + y' - 2y = x \Rightarrow y = -\frac{1}{2}x + \frac{1}{4}$

$y'' + y' - 2y = e^x \Rightarrow y = Ae^x$. $y' = Ae^x(x+1)$, $y'' = Ae^x(x+2)$

$y'' + y' - 2y = Ae^x(2) = 3Ae^x$. $A = \frac{1}{3}$.

$y = \frac{1}{3}xe^x$.

$y'' + y' - 2y = \sin x \Rightarrow y = A\sin x + B\cos x$

$y' = A\cos x - B\sin x$

$y'' = -A\sin x - B\cos x$

$y'' + y' - 2y = (-3A-B)\sin x + (A-3B)\cos x$

$\begin{cases} 3A+B = -1 \\ A = 3B \end{cases} \Rightarrow \begin{cases} A = -\frac{3}{10} \\ B = -\frac{1}{10} \end{cases} \quad y = -\frac{3}{10}\sin x - \frac{1}{10}\cos x$

Step 2. 齐次通解. $y = Ae^x + Be^{-2x}$.

综合. $y = \frac{1}{3}xe^x - \frac{3}{10}\sin x - \frac{1}{10}\cos x + Ae^x + Be^{-2x} - \frac{1}{2}x + \frac{1}{4}$

$y' = \frac{1}{3}e^x(x+1) - \frac{3}{10}\cos x + \frac{1}{10}\sin x + Ae^x - 2Be^{-2x} - \frac{1}{2}$.

$\begin{cases} y(0) = -\frac{1}{10} + A + B - \frac{1}{4} = -\frac{7}{20} \\ y'(0) = \frac{1}{3} - \frac{3}{10} + A - 2B - \frac{1}{2} = \frac{38}{15} \end{cases} \Rightarrow \begin{cases} A+B=0 \\ A-2B=3 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-1 \end{cases}$

$y = \frac{1}{3}xe^x - \frac{3}{10}\sin x - \frac{1}{10}\cos x + e^x - e^{-2x} - \frac{1}{2}x + \frac{1}{4}$.

② $f(x)$ 较复杂, 使用常数变易法.

一般题目限于二阶方程.

Step 1. 先解出齐次通解 $y = Ae^{\alpha x} + Be^{\beta x}$.

Step 2 猜 $y = A(x)e^{\alpha x} + B(x)e^{\beta x}$.

$$y' = \alpha A(x)e^{\alpha x} + \beta B(x)e^{\beta x} + \underline{A'(x)e^{\alpha x} + B'(x)e^{\beta x}}$$

令之为0. 因为有A, B 2个自由度. 这里可以用掉一个. ☆

$$y'' = \alpha^2 A(x)e^{\alpha x} + \beta^2 B(x)e^{\beta x} + \alpha A'(x)e^{\alpha x} + \beta B'(x)e^{\beta x}.$$

$$\text{求解} \begin{cases} ?y'' + ?y' + ?y = f(x) \\ A'(x)e^{\alpha x} + B'(x)e^{\beta x} = 0 \end{cases} \quad \text{即可}$$

Eg. (2025.7). $y'' + 3y' + 2y = \frac{1}{2+e^x}$. (这和作业题 9.6.1. 一样).

Eg. $y'' + 4y = 2 \tan x$.

Step 1. 齐次通解 $y = A \sin 2x + B \cos 2x$.

Step 2. 常数变易 $y = A(x) \sin 2x + B(x) \cos 2x$.

$$y' = 2A(x) \cos 2x - 2B(x) \sin 2x + \underline{(A'(x) \sin 2x + B'(x) \cos 2x)}$$

令之为0

$$y'' = -4B(x) \cos 2x - 4A(x) \sin 2x + 2A'(x) \cos 2x - 2B'(x) \sin 2x$$

$$y'' + 4y = 2A'(x) \cos 2x - 2B'(x) \sin 2x = 2 \tan x$$

$$\Rightarrow \begin{cases} A'(x) \sin 2x + B'(x) \cos 2x = 0 \\ A'(x) \cos 2x - B'(x) \sin 2x = \tan x. \end{cases}$$

$$\Rightarrow A'(x) = -B'(x) \cot 2X.$$

$$\Rightarrow -B'(x) = \tan X \cdot \sin 2X = 2 \sin^2 X$$

$$\Rightarrow B(x) = \int -2 \sin^2 X dX = \sin X \cos X - X.$$

$$A'(x) = 2 \sin^2 X \cdot \frac{\cos 2X}{\sin 2X} = \sin X \cos 2X / \cos X$$

$$A(x) = \ln |\cos X| - \cos^2 X.$$

$$\Rightarrow Y = \sin 2X \cdot \ln |\cos X| - \underbrace{\sin 2X \cos^2 X + \sin X \cos X \cos 2X - X \cos 2X}_{= -\frac{1}{2} \sin 2X}$$

为特解

$$\Rightarrow Y = \sin 2X \cdot \ln |\cos X| - X \cos 2X + A \cos 2X + B \sin 2X.$$

变式 2. Euler 方程. 将一些 $X^n y^{(n)}$ 加起来称为 Euler eqn.

换元 $X = e^t$, $t = \ln X$. 会变成关于 t 的线性方程.

$$\text{Eg. (2023.2). } X^2 y'' - 3X y' + 4y = 0 \quad (X > 0), \quad y(1) = 1, \quad y'(1) = 1.$$

$$\text{换元 } X = e^t, \quad t = \ln X, \quad \frac{dX}{dt} = e^t, \quad \frac{dt}{dX} = e^{-t} = \frac{1}{X}$$

$$Y' = \frac{dy}{dX} = \frac{dy}{dt} \frac{dt}{dX} = \frac{1}{X} \frac{dy}{dt}$$

$$Y'' = \frac{d}{dX} \left(\frac{1}{X} \frac{dy}{dt} \right) = -\frac{1}{X^2} \frac{dy}{dt} + \frac{1}{X} \frac{d^2 y}{dt^2} \frac{dt}{dX} \\ = \frac{1}{X^2} \left(\frac{d^2 y}{dt^2} - \frac{dy}{dt} \right).$$

$$\Rightarrow \underbrace{\frac{d^2 y}{dt^2}}_1 - \underbrace{\frac{dy}{dt}}_{-4} - \underbrace{3 \frac{dy}{dt}}_4 + 4y = 0.$$

$$X^2 - 4X + 4 = 0 \quad \text{特征根} \\ \text{是 } 2 \text{ (二重)}$$

$$\Rightarrow Y = (A + B) e^{2t} = X^2 (A \ln X + B).$$

$$Y' = 2AX \ln X + 2BX + AX.$$

$$\begin{cases} Y(1) = B = 1 \\ Y'(1) = A + 2B = 1 \end{cases} \Rightarrow \begin{cases} A = -1 \\ B = 1 \end{cases}$$

$$\Rightarrow Y = X^2 (1 - \ln X).$$

2. 其它 ODE 相关往年题目.

① 写 Picard 序列.

(2024.5). $y' = x + y^2$. $y(0) = 0$. 写出 Picard 序列 y_1, y_2 .

积分: $y(x) = \int_0^x (s + y^2(s)) ds$.

$$y_1(x) = \int_0^x (s + 0) ds = \frac{1}{2} x^2.$$

$$y_2(x) = \int_0^x (s + \frac{1}{4} s^4) ds = \frac{1}{2} x^2 + \frac{1}{20} x^5.$$

② (2025.6). $xy' - y = (x+y) \ln \frac{x+y}{x}$

这是齐次方程. 令 $y = ux$. ~~$\frac{dy}{dx}$~~

$$y' = u'x + u.$$

$$\Rightarrow x^2 u' + xu - ux = x(1+u) \cdot \ln(1+u).$$

$$\Rightarrow u' = \frac{1}{x} (1+u) \ln(1+u).$$

$$\Rightarrow \frac{du}{(1+u) \ln(1+u)} = \frac{dx}{x} \quad \Rightarrow \ln|\ln(1+u)| = C + \ln|x|.$$

$$\Rightarrow Ce^x = \ln \frac{x+y}{x}. \quad (C \in \mathbb{R}).$$

③ (2021.6). $y' = xy + 3x + 2y + 6$

$$y' = (x+2)(y+3) \quad \Rightarrow \quad \frac{dy}{y+3} = (x+2) dx$$

$$\Rightarrow \ln|y+3| = \frac{1}{2} x^2 + 2x + C. \quad (C \in \mathbb{R}).$$

~~$\Rightarrow$~~ $y =$

3. ODE 解法回顾.

① 分离变量.

$$f(x)dx = g(y)dy \Rightarrow f(x) - g(y) = C.$$

(i) $y' = f(x, y)$. 关于 x, y 齐次. 换元 $y = u x$.

(ii) $y' = f(ax + by + c)$. 换元 $u = ax + by + c$

(iii) $y' = \frac{ax + by + c}{dx + ey + f}$, 若 $\begin{vmatrix} a & b \\ c & d \end{vmatrix} \neq 0$. 平移 $x \leftarrow x - x_0$
 $y \leftarrow y - y_0$.

变成 $(y - y_0)' = y' = \frac{a(x - x_0) + b(y - y_0)}{d(x - x_0) + e(y - y_0)}$

关于 $x - x_0, y - y_0$ 齐次.

若 $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = 0$. 则就是 (ii).

② 积分因子.

$$P(x, y)dx + Q(x, y)dy = 0.$$

(i) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$. 直接积分.

(ii) 否则找 $u(x, y)$. s.t. $\frac{\partial}{\partial y}(P u) = \frac{\partial}{\partial x}(Q u)$.

常见: $u(x), u(y), \frac{1}{x^2 + y^2}, \frac{1}{xy}$.

③ 一阶线性 ODE.

$$y' + P(x)y = Q(x)$$

(i) 齐次 $Q(x) = 0$. $y = C e^{-\int_a^x P(t)dt}$

(ii) 非齐次. 公式法. $y = \left(\int_a^x Q(t) e^{\int_a^t P(s)ds} dt + C \right) e^{-\int_a^x P(t)dt}$

常数变易法: $y = C(x) e^{-\int_a^x P(t)dt}$, 解 $C(x)$.

(iii) Bernoulli 方程 $y' + P(x)y = Q(x)y^\alpha$. ($\alpha \neq 0, 1$)

换元 $z = y^{1-\alpha}$. $\Rightarrow z' + (1-\alpha)P(x)z = Q(x) \cdot (1-\alpha)$.

④ 高阶常系数线性 ODE.

(i) 特征方程

(ii) 非齐次 $\left\{ \begin{array}{l} \text{猜特解 } (P_m(x), e^{\alpha x}, \sin \beta x) \\ \text{常数变易法.} \end{array} \right.$

(iii) Euler 方程. 换元 $x = e^t, t = \ln x$.

(iv) Euler 方程的特征方程.

Eg. $x^2 y'' - 3xy' + 4y = 0$.

代入测试解 x^α .

$$\Rightarrow (\alpha(\alpha-1) - 3\alpha + 4) x^\alpha = 0$$

$$\Rightarrow \alpha^2 - 4\alpha + 4 = 0 \Rightarrow \alpha = 2 \text{ 是二重根}$$

故 $y = x^2 \cdot \underbrace{P_1(\ln x)}_{\text{一次多项式}}.$

一般地, 将 x^α 代入 Euler 方程.

若有 m 重实特征根 α , 则对应基解 $P_{m-1}(\ln x) x^\alpha$.

若有 m 重复特征根 $\alpha \pm \beta i$.

$$\text{由于 } x^{\alpha \pm \beta i} = x^\alpha \cdot x^{\pm \beta i} = x^\alpha \cdot e^{\pm i \cdot \beta \ln x}$$

$$= x^\alpha \cdot (\cos(\beta \ln x) \pm i \sin(\beta \ln x)).$$

因此对应基解 $P_{m-1}(\ln x) x^\alpha \cos(\beta \ln x) + Q_{m-1}(\ln x) x^\alpha \sin(\beta \ln x)$.